

Project Design Phase Proposed Solution Template

Date	15 February 2025
Team ID	
Project Name	AI FAQ Assistant API
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	Support teams spend most of their day answering the same questions, and customers wait for replies. FAQ pages are rarely read, keyword search misses questions asked in different words, and generic chatbots are not tied to company content, so they can give wrong answers. There is no simple, secure API that answers customer questions accurately from a company's own FAQs.
2	Idea / Solution description	AI FAQ Assistant API is a NestJS REST backend. Editors manage FAQs (create, import, categorise, publish). When a customer asks a question, the API finds the most relevant FAQs by meaning (embeddings stored in PostgreSQL with pgvector), then asks an AI model to write a short answer using only those FAQs. Each answer comes with source links and a confidence score. If confidence is low, the API says it does not know and offers to pass the question to a human agent. Feedback and analytics show which questions need new FAQs. Access uses JWT for staff and API keys for client apps, and all endpoints are documented with Swagger.
3	Novelty / Uniqueness	Answers are grounded: the AI sees only approved FAQ text, which reduces made-up answers. Confidence scoring with a "I don't know" fallback and human handoff builds trust. Unanswered-question tracking turns customer questions into new FAQs. Other points: the AI provider can be swapped without changing the rest of the system; Redis caching of repeated questions cuts cost and delay; and each organisation's FAQs are kept separate behind its own API key.
4	Social Impact / Customer Satisfaction	Customers get correct answers in seconds, at any hour, instead of waiting for email. Small businesses can offer good support without a large team. Support staff spend less time on repeat questions and more on problems that need a person. Developers get a ready-made, well-documented API, which shortens their build time. Clear sources and human handoff make AI use more honest and responsible.

5	Business Model (Revenue Model)	Freemium API: a free tier with a limited number of FAQs and answered questions per month; paid plans (Pro and Team) with higher limits, analytics and priority support; usage-based pricing for extra answered questions; enterprise plans with private deployment and custom integrations; optional white-label licensing for helpdesk and website platforms.
6	Scalability of the Solution	Stateless NestJS services scale horizontally behind a load balancer and run in Docker containers. PostgreSQL with a pgvector index handles similarity search, with read replicas when needed. Redis caches answers and holds rate-limit counters. Embedding creation and bulk imports can move to a queue (for example BullMQ) so the API stays fast. The modular design allows the AI module to become a separate service later.